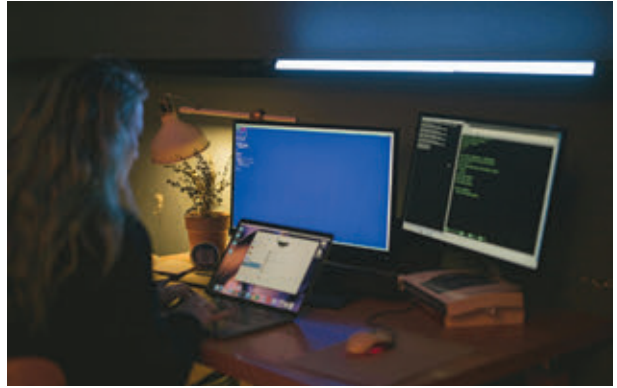




## Setting Up C++

Before you start coding in C++, you need to set up your development environment. Here's how you can get started:

### Installing a Compiler and IDE:

#### 1. Choose a Compiler:

- Popular C++ compilers include GCC (GNU Compiler Collection) for Linux, MinGW for Windows, and Clang for macOS.

#### 2. Install GCC (Linux):

Open a terminal and run the following command:

```
sudo apt update
sudo apt install g++
```

#### 3. Install MinGW (Windows):

- Go to the MinGW website.
- Download the MinGW installer and follow the instructions to install.

#### 4. Install Clang (macOS):

Open a terminal and run the following command:

```
xcode-select --install
```

#### 5. Choose an IDE:

- Popular IDEs for C++ development include Visual Studio, CLion, and Code::Blocks.
- **Visual Studio:** Download from Visual Studio (<https://visualstudio.microsoft.com/>).
- **CLion:** Download from JetBrains (<https://www.jetbrains.com/clion/>).
- **Code::Blocks:** Download from Code::Blocks.

#### 6. Install and Configure Your IDE:

Follow the installation instructions for your chosen IDE. Configure the IDE to use the installed C++ compiler.

## Basic C++ Syntax

Understanding the basic syntax of C++ is essential for writing your first program. Here are some fundamental concepts and constructs:

### Variables and Data Types:

C++ supports various data types, including integers, floats, characters, and booleans. Variables must be declared with a specific type.

```
#include <iostream>
```

```
int main() {
    // Integer
    int age = 25;
```

```
    // Float
    float height = 5.9f;
```

```
    // Character
    char initial = 'A';
```

```
    // Boolean
    bool isStudent = true;
```

```
    std::cout << "Age: " << age << std::endl;
    std::cout << "Height: " << height <<
    std::endl;
```

```
    std::cout << "Initial: " << initial
    << std::endl;
```

```
    std::cout << "Is Student: " <<
    std::boolalpha << isStudent << std::endl;
```

```
    return 0;
```

```
}
```

### Comments:

Comments in C++ can be single-line, starting with `//`, or multi-line, enclosed in `/* */`.

```
// This is a single-line comment
```